Extending the P-graph Formalism

multi-objective ABB, signed-cycle correspondence,
and open generalisations

For the HyMeKo project,
in dialogue with the Friedler P-graph community

2026-05-19

Abstract

The Friedler P-graph formalism (Friedler, Tarján, Huang, Fan 1992) solves the cost-minimising chemical-process-synthesis problem via two structural bounds (inclusion and reachability) over a bipartite material/operating-unit hypergraph. We give a self-contained restatement that exposes the abstract structure independent of the chemical-process domain, then extend the formalism along three axes: (i) multi-objective ABB via positive-weighted scalarisation, with an admissibility-preservation theorem; (ii) a structural correspondence between Friedler ABB and the signed-cycle top-K enumeration ABB developed in this project; and (iii) four open generalisations — hierarchical, time-indexed, stochastic, and categorical — with axiom sketches and the natural next theorems for each. Every result that has shipping code is called out; everything else is sketched at the formal level sufficient for the next iteration to be written.

Contents

1 Preliminaries and notation

Definition 1 (P-graph)**.** A *P-graph* is a 6-tuple $\mathcal{P} = (M, O, \mathrm{in}(\cdot), \mathrm{out}(\cdot), R, P)$ where:

- M is a finite set of *materials*;
- O is a finite set of *operating units*, disjoint from M;

- $\mathrm{in}(\cdot) : O \to 2^M$ assigns to each unit its *input materials* (consumed);
- $\mathrm{out}(\cdot) : O \to 2^M$ assigns to each unit its *output materials* (produced);
- $R \subseteq M$ is the *raw-material* set (feedstocks);
- $P \subseteq M$ is the *product-demand* set (required outputs).

The directed bipartite incidence graph is $E_{\mathrm{dir}} = \{(m,u) : u \in O,\ m \in \mathrm{in}(u)\} \cup \{(u,m) : u \in O,\ m \in \mathrm{out}(u)\}$.

Definition 2 (Producibility closure)**.** Given $O' \subseteq O$, the *producibility closure* of the raw set under O' is the least fixed point

$$C_R(O') \;=\; \mathrm{lfp}\Big[\, X \;\mapsto\; R \cup \{\, m \in M \;:\; \exists u \in O',\ \mathrm{in}(u) \subseteq X,\ m \in \mathrm{out}(u) \,\}\Big].$$

Operationally, $C_R(O')$ is built by iteratively adding the outputs of every unit whose inputs are already producible from R. The iteration converges in $O(|O|)$ rounds.

Definition 3 (Feasibility)**.** A subset $O' \subseteq O$ is a *solution structure* if:

(a) **Input consistency:** for every $u \in O'$, $\mathrm{in}(u) \subseteq C_R(O')$.

(b) **Demand:** $P \subseteq C_R(O')$.

(c) **Strict no-excess (optional):** every produced material not in $R \cup P$ is consumed by some $u \in O'$.

The set of all solution structures is denoted $\mathfrak{S}(\mathcal{P})$.

Definition 4 (Cost interface)**.** A *cost interface* on $\mathcal{P}$ is a non-negative function $c : O \to \mathbb{R}_{\geq 0}$. The *total cost* of $O' \subseteq O$ is $\mathrm{tc}_c(O') = \sum_{u \in O'} c(u)$. The *cost-minimising synthesis problem* is

$$\min_{O' \in \mathfrak{S}(\mathcal{P})} \ \mathrm{tc}_c(O'). \tag{1}$$

2 The classical Friedler axioms

We restate the five axioms with explicit logical content; the 1992 paper's prose is slightly compressed.

Axiom A1 (Product-membership)**.** Every required product $p \in P$ is a material: $P \subseteq M$.

Axiom A2 (Raw-reachability)**.** Every material in M is either raw or producible by some unit chain from raws: $M \subseteq C_R(O)$.

Axiom A3 (Unit-catalogue closure)**.** Every $u \in O$ has $\mathrm{in}(u) \cup \mathrm{out}(u) \subseteq M$ and $\mathrm{in}(u) \cap \mathrm{out}(u) = \emptyset$.

Axiom A4 (Connectedness)**.** For every $u \in O$, both $|\mathrm{in}(u)| \geq 1$ and $|\mathrm{out}(u)| \geq 1$ (no orphan units).

Axiom A5 (No-cycle on materials)**.** The directed bipartite incidence graph E_{dir} has no directed cycle that visits a material twice. (Equivalently, E_{dir} is a DAG when contracted on materials.)

Remark 5. **A1** and **A2** are well-formedness conditions on the problem statement; **A3**–**A5** are *structural feasibility* preconditions that any P-graph admitted to MSG must satisfy. Stage P-mo's lowering enforces **A3**–**A4** at parse time; **A5** is enforced by the bipartite-schema invariant in HyMeKo.

3 MSG, SSG, ABB as operators on the subset lattice

Definition 6 (Maximal structure $\mathrm{MSG}(\mathcal{P})$)**.** The *maximal structure* is $\mathrm{MSG}(\mathcal{P}) = \bigcup_{O' \in \mathfrak{S}(\mathcal{P})} O'$. Equivalently (Friedler 1992), MSG is the largest O' that survives the fixed-point iteration of forward-feasibility (drop u whose inputs are not in $C_R(O')$) and backward-utility (drop u whose outputs are not in the union of inputs of remaining units $\cup P$) trims.

Definition 7 (SSG)**.** SSG($\mathcal{P}$) enumerates every $O' \subseteq \mathrm{MSG}(\mathcal{P})$ that satisfies feasibility (Def. 3). For $|\mathrm{MSG}(\mathcal{P})| > 30$, SSG is impractical and ABB is the right tool.

Definition 8 (ABB)**.** ABB($\mathcal{P}, c$) returns $\arg\min_{O' \in \mathfrak{S}(\mathcal{P})} \mathrm{tc}_c(O')$ via depth-first include/exclude branching on a fixed ordering of $\mathrm{MSG}(\mathcal{P})$, with two bounds:

B1 Inclusion bound. Let s.cost be the partial sum at the current node and s.best the cost of the current incumbent. If $s.\mathrm{cost} \geq s.\mathrm{best}$, prune.

B2 Reachability bound. Let $\widehat{O} = \mathrm{included} \cup \mathrm{undecided}$. If $C_R(\widehat{O}) \not\supseteq P$, prune.

Branching is include-before-exclude so the inclusion bound has an incumbent to compare against as early as possible.

Proposition 9 (ABB correctness)**.** *If both **B1** and **B2** are admissible (prune only nodes whose subtree cannot improve the incumbent), then ABB($\mathcal{P}, c$) returns an optimal feasible solution structure, or reports infeasibility if $\mathfrak{S}(\mathcal{P}) = \emptyset$.*

Proof sketch. **B1** is admissible because every inclusion adds a non-negative cost, so the partial cost lower-bounds the descendant's full cost. **B2** is admissible because $C_R(\widehat{O}) \supseteq C_R(O')$ for every $O' \subseteq \widehat{O}$ (monotonicity of C_R in its argument), so if $C_R(\widehat{O}) \not\supseteq P$, no descendant subset can be feasible. Since both bounds preserve every optimum, the DFS visits at least one optimum and returns the minimum. $\square$

4 Multi-objective extension: positive-weighted scalarisation

We now lift the cost interface from a scalar function $c : O \to \mathbb{R}_{\geq 0}$ to a D-dimensional vector function $\boldsymbol{c} : O \to \mathbb{R}^D_{\geq 0}$ across named dimensions (e.g. CAPEX, OPEX, CO_2, H_2O).

Definition 10 (Multi-objective cost interface)**.** A *D-dimensional cost interface* on $\mathcal{P}$ is a function $\boldsymbol{c} : O \to \mathbb{R}^D_{\geq 0}$, written $\boldsymbol{c}(u) = (c_1(u), \ldots, c_D(u))$. Given a user-supplied *weight vector* $\boldsymbol{w} \in \mathbb{R}^D_{\geq 0}$, the *weighted-sum cost* of $u \in O$ is $c_{\boldsymbol{w}}(u) = \langle \boldsymbol{w}, \boldsymbol{c}(u) \rangle = \sum_{d=1}^{D} w_d \cdot c_d(u)$, and the weighted total cost of $O' \subseteq O$ is $\mathrm{tc}_{\boldsymbol{w}}(O') = \sum_{u \in O'} c_{\boldsymbol{w}}(u)$.

Theorem 11 (Admissibility preservation under positive-weighted scalarisation)**.** *For every $\boldsymbol{w} \in \mathbb{R}^D_{\geq 0}$ and every multi-objective cost interface $\boldsymbol{c} : O \to \mathbb{R}^D_{\geq 0}$, the ABB inclusion bound **B1** restated as*

$$\textit{prune when} \sum_{u \in \mathrm{included}} c_{\boldsymbol{w}}(u) \;\geq\; s.\mathrm{best}_{\boldsymbol{w}}$$

is admissible, and Proposition 9 extends: ABB($\mathcal{P}, \boldsymbol{c}, \boldsymbol{w}$) returns $\arg\min_{O' \in \mathfrak{S}(\mathcal{P})} \mathrm{tc}_{\boldsymbol{w}}(O')$.

Proof. Admissibility of **B1** requires that the partial weighted cost be a lower bound on every descendant's full weighted cost. Since $w_d \geq 0$ for every d and $c_d(u) \geq 0$ for every d, u, every inclusion-step term $c_{\boldsymbol{w}}(u) = \sum_d w_d c_d(u)$ is non-negative. The partial sum is therefore monotone-non-decreasing along the include-edge of the search tree, so the partial weighted cost lower-bounds the descendant's full weighted cost. The reachability bound **B2** is structural and does not depend on the cost interface; it is unaffected. The rest of Prop. 9 applies verbatim. $\square$

Remark 12 (Why non-negativity matters)**.** Theorem 11 fails for general $\boldsymbol{w} \in \mathbb{R}^D$. If some $w_d < 0$, including a unit can decrease the partial cost, and the inclusion bound no longer holds. A *benefit term* (negative weight, e.g. revenue) can be made admissible by treating it separately — subtract the maximum possible benefit of the remaining subtree from the bound, see Section 6.

Corollary 13 (Implementation status)**.** *The reference implementation in `hymeko_pgraph::abb` (commit Stage P-mo, 2026-05-19) realises Theorem 11 verbatim. The new ABBOptions field `cost_weights: Option<Vec<f64>>` dispatches between the scalar path (Friedler 1992, byte-identical) and the weighted-sum path. The methanol-synthesis worked example (`data/pgraph/methanol_synthesis.hymeko`) demonstrates four structurally distinct optima from five weight regimes, including a non-obvious hybrid topology that no single-criterion regime selects.*

5 The signed-cycle correspondence

Beyond chemical-process synthesis, this project has independently applied ABB-style branch-and-bound to a structurally different problem: top-K enumeration of signed cycles in a signed graph for graph-neural-network training. We now show these two ABB instances are not analogues but *the same abstract algorithm.*

Definition 14 (Top-K signed-cycle enumeration)**.** Given a signed graph $G = (V, E, \sigma : E \to \{-1, +1\})$, a cycle length k, a target heap size K, and a per-cycle score function $s : \mathrm{Cyc}_k(G) \to \mathbb{R}$, the *top-$K$ enumeration problem* returns the K closed cycles of length k with the largest scores. ABB uses a *bounded scorer trait* $(s, \bar{s})$, where $\bar{s}(n_{\mathrm{neg}}, k_{\mathrm{rem}}, k)$ is an upper bound on the score of any closed extension of a partial path with n_{neg} negative edges so far and k_{rem} remaining edges to close, and prunes when $\bar{s} \le$ heap.peek_min.

Theorem 15 (Cycle-PSE correspondence)**.** *The cycle-enumeration ABB (Def. 14) and the P-graph ABB (Section 3) are both instances of the following abstract algorithm. Let $\mathcal{T}$ be a finite search tree with leaves $\mathcal{L}$; let $\mathrm{obj} : \mathcal{L} \to \mathbb{R}$ be an objective; let $\overline{\mathrm{obj}} : \mathrm{nodes}(\mathcal{T}) \to \mathbb{R}$ be an admissible upper bound (or lower bound for minimisation) on the optimum over each subtree. Then* ***Algorithm A1*** *[DFS-with-bound] visits at least one optimum and returns it in time $O(|\mathcal{T}_{\overline{\mathrm{obj}}\text{-}feasible}|)$, where $|\mathcal{T}_{\overline{\mathrm{obj}}\text{-}feasible}|$ counts the nodes whose bound does not allow pruning.*

Proof sketch. In the cycle case, $\mathcal{L}$ is the set of all closed cycles of length k from a chosen start vertex, obj is the per-cycle score, and the bound is the BoundedScorer's `upper_bound` method. In the PSE case, $\mathcal{L}$ is the set of feasible solution structures, obj is the (possibly weighted) cost, and the bound is the partial-cost-plus-zero-future = partial-cost itself (plus the structural reachability gate). Both fit Algorithm A1 with $\mathcal{T}$ being a binary include/exclude tree (PSE) or a $|V|$-way edge-extend tree (cycles). □

Remark 16. The shared abstract algorithm explains why the implementation patterns parallel each other so closely. The HyMeKo codebase duplicates structural code across the two ABB instances; a plausible refactor (not in scope here) would extract a `trait BoundedSearch` that both inhabit.

6 Open generalisations

We sketch four extensions, each with the natural next axiom or theorem and a pointer to the next implementation step.

6.1 Pareto-front enumeration

For a multi-objective P-graph $(\mathcal{P}, \boldsymbol{c}, D)$, the *Pareto front* $\mathcal{F} \subseteq \mathfrak{S}(\mathcal{P})$ is the set of feasible structures O' such that no other feasible O'' satisfies $c_d(O'') \le c_d(O')$ for all d with strict inequality for at least one. Theorem 11 gives *one* Pareto-optimal point per $\boldsymbol{w}$; sweeping $\boldsymbol{w}$ over the unit simplex enumerates the whole front.

Proposition 17 (Simplex-grid completeness for $D \le 3$)**.** *For $D \le 3$ and bounded integer cost vectors, the Pareto front $\mathcal{F}$ is finite, and a simplex-grid sweep of $\boldsymbol{w}$ at resolution $1/N$ recovers every Pareto-optimal structure for sufficiently large N (depending on the gaps between adjacent Pareto points).*

Proof sketch. The Pareto front of finitely many $\mathbb{R}^D_{\ge 0}$ points is itself finite. For each Pareto point $O^\star$, the cone of weights $\boldsymbol{w}$ for which $O^\star$ is the weighted optimum is open and non-empty (otherwise $O^\star$ is dominated). A fine-enough simplex grid intersects every such cone. □

Implementation step: ~ 100 LOC, builds on the existing SSG enumeration loop. For $D \ge 4$ the simplex grid is exponential in D and sampling becomes more efficient than grid; adaptive (ϵ-constraint or weighted-L^2-Tchebycheff) methods take over.

6.2 Hierarchical P-graphs

A natural generalisation models each operating unit as itself a P-graph (a sub-plant). Formally:

Definition 18 (Hierarchical P-graph)**.** A *hierarchical P-graph* is a tuple $\mathcal{H} = (M, O, \mathrm{in}(\cdot), \mathrm{out}(\cdot), R, P, \mathrm{sub})$ where $\mathrm{sub} : O \to \mathfrak{P}_{\mathrm{hier}}$ is a (possibly partial) function assigning to each unit a sub-plant (itself a hierarchical P-graph). The flat unrolling of $\mathcal{H}$ substitutes each $u \in O$ with $\mathrm{sub}(u)$, plumbing the boundary materials through.

Proposition 19 (Hierarchical ABB)**.** *ABB on a hierarchical P-graph $\mathcal{H}$ reduces to nested ABB calls: solve each sub-plant for the cost it contributes to its parent unit, then run parent-level ABB with the cached sub-plant costs. The reachability bound becomes a* nested closure*: a parent unit is reachable iff its sub-plant has a feasible structure under the parent's input materials.*

Implementation step: sub-plant caching plus a recursive ABB entry. The bound for the nested closure requires care; the natural extension is to compute the sub-plant's minimum cost over its input/output interface as the parent-level cost.

6.3 Time-indexed P-graphs

Real plants operate in time. We add an explicit time index.

Definition 20 (Time-indexed P-graph)**.** A *time-indexed P-graph* over a finite horizon $T = \{0, 1, \ldots, H\}$ is a tuple $(\mathcal{P}, \tau_{\mathrm{in}}, \tau_{\mathrm{out}}, S)$ where:

- $\tau_{\mathrm{in}}, \tau_{\mathrm{out}} : O \to \mathbb{Z}_{\geq 0}$ are per-unit consumption and production latencies;
- $S \subseteq M$ is a set of *storable* materials (others must be consumed in the same time-step as produced).

A *time-indexed solution structure* is a function $\Sigma : T \to 2^O$ assigning to each time-step the set of units active at that step, subject to a time-extended feasibility: inputs at time t must be available from raws or from outputs produced no later than $t - \tau_{\mathrm{in}}(u)$, and outputs of non-storable materials must be consumed at exactly $t + \tau_{\mathrm{out}}(u)$.

Axiom A1 (Time-monotonicity / A6)**.** For every storable material $m \in S$ and every time-step t, the available inventory of m at t is a non-decreasing function of the production schedule up to t. (Production is irreversible; no negative inventory.)

Implementation step: the rolling-horizon ABB takes $\Sigma : T \to 2^O$ as the lattice, with the reachability bound generalised to per-time-step closure under inventory dynamics. This is non-trivial work; the natural prototype is $H \leq 24$ hourly steps on the methanol example.

6.4 Stochastic P-graphs

Real plants face uncertain feedstock costs, demand, and equipment availability. We treat the cost interface as random.

Definition 21 (Stochastic P-graph)**.** A *stochastic P-graph* is a tuple $(\mathcal{P}, \mathbb{P})$ where $\mathbb{P}$ is a probability measure on a sample space Ω and the cost interface is a random function $c : O \times \Omega \to \mathbb{R}_{\geq 0}$. The decision variable is the structure O'; the realisation $\omega \in \Omega$ is drawn *after* the decision. The *risk-neutral* objective is $\min_{O'} \mathbb{E}_\omega[\mathrm{tc}_c(O', \omega)]$. The *conditional-value-at-risk* (CVaR) objective at level $\alpha \in (0, 1]$ is $\min_{O'} \mathrm{CVaR}_\alpha[\mathrm{tc}_c(O', \omega)]$.

Axiom A2 (Risk-coherence / A7)**.** The risk functional $\rho : L^1(\Omega, \mathbb{P}) \to \mathbb{R}$ employed in the stochastic objective is *coherent*: monotone, sub-additive, positively homogeneous, and translation-invariant (Artzner et al. 1999). $\mathbb{E}$ and CVaR_α both satisfy A7.

Proposition 22 (Sample-average approximation)**.** *For coherent ρ and a sample $\omega_1, \ldots, \omega_N$ drawn i.i.d. from $\mathbb{P}$, the sample-average ABB*

$$\min_{O'} \rho(\{\mathrm{tc}_c(O', \omega_n)\}_{n=1}^N)$$

satisfies an admissibility-preservation theorem analogous to Theorem 11, because ρ being monotone and translation-invariant preserves the cost-monotonicity of the inclusion bound.

Implementation step: per-realisation cost vectors as multi-objective cost vectors of size $D = N$; the weight vector encodes the risk functional. CVaR_α corresponds to taking the largest $\lceil N\alpha \rceil$ realisations' average; this can be re-cast as a weighted-sum with a structural weight constraint, fitting Theorem 11 after a permutation step.

6.5 Categorical formulation

The bipartite structure of a P-graph and the compositional nature of nested sub-plants suggest a categorical view. Sketch only:

Definition 23 (**PGraph**)**.** Let **PGraph** be the category whose objects are finite sets of materials (with raw/product designations) and whose morphisms $\phi : M_R \to M_P$ are P-graphs from $R = M_R$ to $P = M_P$. Composition $\phi \circ \psi$ glues two P-graphs by identifying the product set of ψ with the raw set of ϕ. The identity morphism is the empty P-graph ($O = \emptyset$, $R = P$).

Proposition 24 (**PGraph** is symmetric monoidal)**.** *The disjoint union of P-graphs equips* **PGraph** *with a symmetric monoidal structure* $\otimes : \mathbf{PGraph} \times \mathbf{PGraph} \to \mathbf{PGraph}$, *with unit* $I = (\emptyset, \emptyset, \emptyset, \emptyset)$. *Recycle loops correspond to the trace operator* $\mathrm{Tr}^X : \mathbf{PGraph}(A \otimes X, B \otimes X) \to \mathbf{PGraph}(A, B)$, *making* **PGraph** *a traced symmetric monoidal category.*

Remark 25. The categorical view connects P-graphs to other formalisms in the SMC ecosystem (Petri nets, signal-flow graphs, string diagrams). The trace structure formalises recycle loops as morphism feedback; the multi-objective cost functor sends **PGraph** to $\mathbf{Lat}^D$ (the category of D-component lattices), with ABB as the value-recovery functor's adjoint. This connection is exploratory and not yet implemented.

Implementation step: none in the current codebase. The categorical framing is a paper-only generalisation that would inform a future redesign of the IR shared between ABB instances.

7 Implementation status and roadmap

Extension	Section	Status	Estimated LOC
Scalar ABB (Friedler 1992)	3	shipped	—
Multi-objective ABB (Thm. 11)	4	shipped Stage P-mo 2026-05-19	~ 120
Cycle-PSE correspondence (Thm. 15)	5	shipped (both ends)	—
Pareto-front enumeration (Prop. 17)	6.1	queued	~ 100
Hierarchical P-graphs (Prop. 19)	6	queued	~ 300
Time-indexed P-graphs (**A6**)	6	sketched	~ 800
Stochastic P-graphs (**A7**, Prop. 22)	6	sketched	~ 400
Categorical **PGraph**	6	exploratory	—

What the formalism is actually used for in this project.

1. Process synthesis on the canonical HDA example (test fixture).
2. Multi-objective methanol synthesis (*this report's worked example*).
3. Top-K signed-cycle enumeration in HSiKAN (graph-neural-net training); shares the abstract Algorithm A1.
4. Graph-property-conditional axiom selection for signed-link prediction; the regime-conditional rule is structurally an ABB-style bound choice (which axiom to enforce given the graph's balance ratio).

8 Bottom line

The Friedler 1992 P-graph formalism is robust enough to bear serious extensions. We've shipped two: the multi-objective ABB with admissibility preservation (Theorem 11) and the structural correspondence to signed-cycle enumeration (Theorem 15). Four further generalisations (Pareto-front, hierarchical, time-indexed, stochastic) are formally sketched with explicit axioms or theorems and a clear implementation step for each. A categorical formulation hints at a deeper unification but is exploratory.

The formalism's value for the PSE community goes beyond its original chemical-process-synthesis domain: any combinatorial decision problem where (a) the search space is the subset lattice of a finite catalogue, (b) feasibility is closed under reachability of a target set, and (c) the objective is non-negatively weighted-additive over the catalogue, is an instance of P-graph ABB. The cycle-enumeration application is a proof of concept that this generality is operational. The next big opportunities — time-indexed plant scheduling, stochastic feedstock procurement, hierarchical sub-plant nesting — are all within the formal envelope established here.

References to the implementation. Plan `docs/plans/2026-05-19-pgraph-multi-objective/`; implementation in `hymeko_pgraph/src/{lowering,abb,dump}.rs`; 9 new tests in `hymeko_pgraph/tests/multi_objective.rs`; worked example in `data/pgraph/methanol_synthesis.hymeko`; companion 5-page demo PDF in `reports/2026-05-19-pgraph-multi-objective.pdf`. The top-K signed-cycle ABB is implemented in `hymeko_graph/src/topk_cycles.rs`; see `reports/2026-05-10-abb-global-topk.md` for the 25.06× Epinions benchmark.

This document is intended as input for an upcoming paper "P-graph methodology beyond PSE: axiom-feasibility for sustainable combinatorial optimisation" (working title) to be discussed with the Friedler community, including Jean Pimentel. Feedback on the axiomatic statements, the categorical framing, and the priority among the open generalisations is welcome.